

Realizability

Suppose I am given a poset, and asked "Is this the face poset of a polytope?"

Necessary conditions:

- Single maximal element
- Lattice after adding minimal element
- Ranked/graded

2 dimensions.



3 dimensions.

faces } cycles of the graph
 } whose deletion do not disconnect
 } the graph
edges } must form a graph
vertices } graph must be planar
 } and 3-connected

Let P be a ranked poset with ranks $0, \dots, d$.

For each rank 0 element v , assign a vector $x^v = (x_1^v, x_2^v, \dots, x_d^v)$ of undetermined variables.

For each element F of P , assign a vector $y^F = (y_1^F, y_2^F, \dots, y_d^F)$ and real number b^F . Want

$$x^v \cdot y^F = b^F \quad \text{if } v \in F \quad \left\{ \begin{array}{l} v \text{ is rank } 0 \\ v \leq F \end{array} \right.$$

$$x^v \cdot y^F < b^F \quad \text{if } v \notin F$$

Want to know if $\exists$ such x^U, y^F, b^F .

The **existential theory of the reals** is the problem of determining whether a system of polynomial inequalities has a solution. It is known to be **NP-hard**.

Thm (Mriev): The realizability problem for polytypes is complete for the existential theory of the reals.

In other words, every instance of an existential theory of reals problem can be reduced in polynomial time to an instance of the realizability problem for polytypes.

Thm (Richter-Gebert): The realizability problem for 4-polytopes is complete for the existential theory of the reals.

Given a graph, is it the 1-skeleton of a polytope?

Also $\exists$ complete for $\exists \text{PR}$

Given a graph, is it the 1-skeleton of a 4-polytope?

Unknown

Given a poset, is it the face poset of a
regularly polytope?

Complete for $\exists \text{PR}$

Given a finite set of points $v_1, \dots, v_d$ in $\mathbb{R}^d$,
consider the sign pattern of the maximal minors of

$$\begin{pmatrix} 1 & 1 & \dots & 1 \\ v_1 & v_2 & \dots & v_n \end{pmatrix}$$

This is the **oriented matroid** of $v_1, \dots, v_d$.

The **realization space** of the oriented matroid is
the space of all $(v_1, \dots, v_d)$ which give this sign
pattern.

Thm (Mäev): The realization space of an oriented matroid can have the homotopy type of any semialgebraic set.

solution set to a system
of polynomial inequalities

Note: The realization space of $C(d, n)$ is contractible. (totally positive Grassmannian)